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Complex permittivity measurements of low-loss microwave ceramics employing higher order quasi TE_{0np} modes excited in a cylindrical dielectric sample

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Abstract

Higher order quasi TE_{0np} modes have been used for precise measurements of permittivity and dielectric loss tangent of low-loss dielectric ceramics versus frequency in the microwave frequency range. It has been shown that, using this technique, dielectrics having arbitrary permittivity value and dielectric loss tangent values in the range 10^{-6} – 10^{-2} can be measured with relative uncertainties below 0.5% for real permittivity and below 5% for dielectric losses. Several dielectric materials have been measured including single-crystal quartz, single-crystal sapphire and the low-loss microwave ceramics BMT, BZT and BZNT.

Keywords: dielectric resonators, dielectric properties, tantalates

1. Introduction

One of the most frequently used techniques for measuring the complex permittivity of low-loss dielectric materials is the Hakki–Coleman type dielectric resonator and its modifications [1–3]. The original Hakki–Coleman technique is usually used to measure real permittivity while dielectric losses are more often measured with the $TE_{01\delta}$ mode resonator method [4, 5], mainly because of its small (very often negligible) parasitic losses. The $TE_{01\delta}$ mode resonator technique is commonly used for dielectric loss measurements of high-permittivity low-loss dielectric ceramics [4]. One reason why real permittivity is rarely measured using this technique is the difficulties in accurate electromagnetic modeling of the $TE_{01\delta}$ mode resonator structure.

Both Hakki–Coleman and $TE_{01\delta}$ mode resonator techniques are restricted to measurements at a single frequency. In principle it is possible to extend the range of measurement frequency using several samples made of the same material, but with increasing frequency the dimensions of the samples become very small and measurement uncertainties

increase. This is the main reason why both the above-mentioned techniques are rarely used for measurements at frequencies above 10 GHz, especially for high permittivity materials.

Measurements at higher microwave frequencies using one dielectric specimen are possible by employing higher order resonant modes that can be excited in a dielectric resonator created by the sample under test. One of the most accurate techniques of this kind is the whispering gallery mode (WGM) technique [6, 7] that employs higher order hybrid modes excited in cylindrical dielectric specimens. The WGM technique is especially useful for measurements of extremely low-loss materials at cryogenic temperatures (losses as low as 10^{-9} have been measured with this technique [6]) but it is difficult to use for measurements of materials with dielectric loss tangents in the range 10^{-4} – 10^{-2} due to mode overlapping, problems with excitations of low Q -factor WGM modes, and problems with mode identification.

In this paper we propose the implementation of higher order TE_{0np} modes that can be used for simultaneous determination of permittivity and dielectric loss tangent

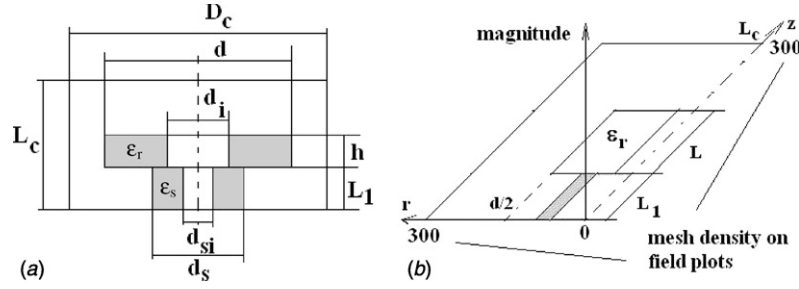


Figure 1. Resonant structure used for measurements of the complex permittivity employing higher order quasi TE_{0np} modes. (a) Sketch of resonant structure; (b) system of coordinates used for visualization of electromagnetic fields.

of isotropic materials at the frequencies of millimetre waves.

2. Theory

The resonant structure proposed for measurements is shown in figure 1(a). Accurate complex permittivity determination using quasi TE_{0np} modes requires numerical electromagnetic computations. For the structure shown in figure 1(a) this can be done by employing one of several rigorous techniques, e.g. finite element, finite difference or mode matching. For all electromagnetic computations performed in this paper we have used the Galerkin–Rayleigh–Ritz (G-R-R) method described, e.g., in [8–10]. The accuracy of numerical evaluation of resonant frequency for the dominant quasi TE_{011} mode employing G-R-R and mode-matching methods has been already studied [10] and it has been proved that computational errors in most practical cases are smaller than experimental ones. As one may expect, convergence of numerical computations of resonant frequencies for higher order quasi TE_{0np} modes will be worse than for the dominant quasi TE_{011} mode; therefore it is important to establish error bounds for the modes of interest. In practice it is important to identify only the first few modes belonging to the quasi TE mode family because experimental identification of proper modes becomes extremely difficult with increasing frequency due to the increase in the density of spurious modes. Our G-R-R software for quasi TE_{0np} mode analysis uses 325 basis functions of the empty cavity and allows simultaneous analysis of several modes and identification of their radial and axial mode indices through a graphical interface that visualizes electromagnetic fields over the resonator cross section shown in figure 1(b).

Assessment of the convergence and accuracy of numerical computations of the TE_{0np} mode resonant frequencies employing the G-R-R technique has been performed in two steps. In the first step the special case of the resonant structure for which ‘exact’ numerical solutions are available has been studied. This takes place when $h = L_c$ for the structure shown in figure 1(a). Results of computations for the first eight TE_{0np} modes are shown in table 1 (modes are presented in order of increasing frequency and their number in the TE mode frequency spectrum is given by the symbol N). ‘Exact’ resonant frequency values have been obtained as the roots of the appropriate transcendental equation [11] that exists for the simplified resonant structure with $h = L_c$. Results in table 1 show that resonant frequencies computed by the G-R-R

Table 1. Computational errors of the resonant frequencies of several TE_{0np} modes for the structure for which exact solutions exist. $D_c = 9.4$ mm, $L_c = 4.12$ mm, $d = 4.12$ mm, $h = 4.12$ mm, $\epsilon_r = 24.5$.

N	n	p	f (GHz) (exact)	f (GHz) Galerkin– Rayleigh–Ritz	Error
1	1	1	14.918	14.918	$<10^{-4}$
2	1	2	20.479	20.479	$<10^{-4}$
3	1	3	26.707	26.707	$<10^{-4}$
4	2	1	27.521	27.521	$<10^{-4}$
5	2	2	31.278	31.279	$<10^{-4}$
6	2	3	35.994	35.996	$<10^{-4}$
7	3	2	44.097	44.103	1.4×10^{-4}
8	3	3	47.722	47.730	1.7×10^{-4}

method provide an upper bound for exact resonant frequencies and that computational errors increase with increasing mode number. When the height of the sample is smaller than the height of the cavity exact solutions are not available. In such a case assessment of the accuracy of any numerical technique is only possible by intercomparisons with other techniques or with experiments. For the dominant quasi TE_{0np} mode the matching technique is complementary to the G-R-R method, providing a lower bound for the resonant frequency. By employing both techniques simultaneously one can obtain rigorous error bounds for the resonant frequency or for permittivity.

Results of such intercomparisons are shown in figure 2. It is seen that computational errors increase for small samples (with increasing aspect ratio), but even for samples having volumes as small as $1/125$ of the cavity volume (aspect ratio equal to 5) the difference between results obtained by the two techniques is smaller than 0.125%. This value is smaller than the measurement uncertainties associated with other sources (e.g. dimensional uncertainties).

As has been already shown for a simple resonant structure, the computational error increases for higher order modes. Results of computations of the resonant frequencies for the first ten TE_{0np} modes of the structures having aspect ratios 2 and 4 and permittivities 10 and 100 employing the G-R-R and mode-matching techniques are shown in figure 3. Magnetic field plots for the first nine modes are shown in figure 4 and allow identification of the modes.

One can see that computational errors are below 0.1% for most of the quasi TE_{0n1} modes if the aspect ratio of the cavity to the sample <4 and permittivity <100 . Computations performed for permittivities >100 have shown similar convergence to those for permittivity equal to 100

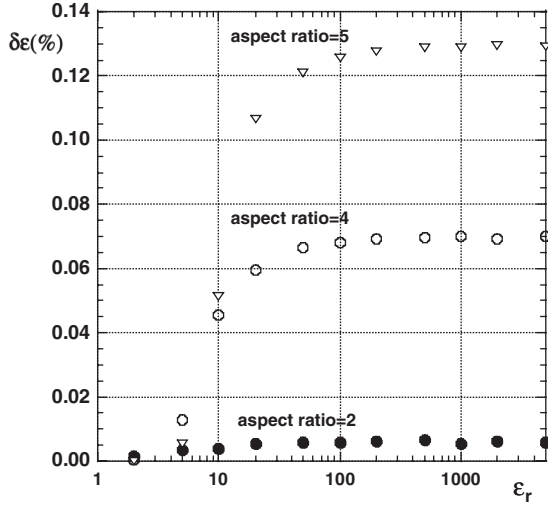


Figure 2. Computational error of real permittivity (difference between results obtained by mode-matching and Rayleigh–Ritz techniques) employing the quasi TE₀₁₁ mode versus permittivity for different aspect ratio defined as $D_c/d = L_c/h$. Cavity $D_c/L_c = 2$, $L_1 = (L_c - h)/2$.

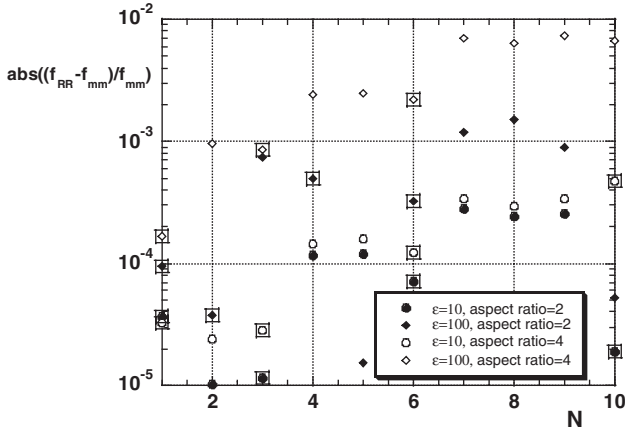


Figure 3. Computational error of resonant frequency (difference between results obtained by mode-matching and Rayleigh–Ritz techniques) employing quasi TE_{0np} modes versus permittivity for different aspect ratios $D_c/d = L_c/h$. Results for quasi TE_{0n1} modes are enclosed in squares. Cavity $D_c/L_c = 2$, $L_1 = (L_c - h)/2$.

(which is in agreement with results shown in figure 2). Therefore if aspect ratio of the cavity to the sample < 4 then TE_{0n1} modes can be used for precise determination of real permittivity.

In contrast to the tight bounds of the relative permittivity (or resonant frequency), it is impossible to obtain corresponding bounds for the dielectric loss tangent. The dielectric loss tangent of the sample under test, $\tan \delta$, is always evaluated from the following formula:

$$\tan \delta = (Q_u^{-1} - p_{es} \tan \delta_s - R_s/G)/p_{er} \quad (1)$$

where Q_u is the measured unloaded Q -factor of a specific quasi TE_{0np} mode, $\tan \delta_s$ is the dielectric loss tangent of the dielectric support, R_s is the surface resistance of cavity walls at the frequency corresponding to the resonant frequency of a specific quasi TE_{0np} mode, G is the geometric factor defined

Table 2. Resonant frequencies, electric energy filling factors and Q -factor due to conductor losses for the resonant structure with $D_c = 20$ mm, $R_c = 30$ mΩ at 10 GHz, $D_c/h = L_c/h = 4$, $D_c/L_c = 2$, $\epsilon_r = 100$, $\epsilon_s = 1$.

N	f (GHz)	p_e	Q_c	Mode designation
1	6.23943	0.98733	174663	quasi TE ₀₁₁
2	10.0699	0.99106	878548	quasi TE ₀₁₂
3	11.7310	0.98730	799251	quasi TE ₀₂₁
4	14.8738	0.98863	1180500	quasi TE ₀₁₃
5	14.7710	0.99454	3735217	quasi TE ₀₂₂
6	17.4230	0.98198	917645	quasi TE ₀₃₁

as

$$G = \frac{\omega \iiint_V \mu_0 |\mathbf{H}|^2 dv}{\oint_S |\mathbf{H}_t|^2 ds} \quad (2)$$

and p_{es} and p_{er} are the electric energy filling factors for the dielectric support and the sample that are defined as

$$p_{es} = \frac{W_{Es}}{W_{ET}} = \frac{\iiint_{V_s} \epsilon_s \mathbf{E} \cdot \mathbf{E} dv}{\iiint_V \epsilon(v) \mathbf{E} \cdot \mathbf{E} dv} \quad (3)$$

$$p_{er} = \frac{W_{Er}}{W_{ET}} = \frac{\iiint_{V_r} \epsilon_r \mathbf{E} \cdot \mathbf{E} dv}{\iiint_V \epsilon(v) \mathbf{E} \cdot \mathbf{E} dv} \quad (4)$$

where W_{ET} is the total electric energy stored in the resonant structure, W_{Es} is the electric energy stored in the dielectric support, W_{Er} is the electric energy stored in the sample under test, V is the total volume of the resonant structure, V_s is the volume of the dielectric support, V_r is the volume of the sample under test, S is the internal metal surface of the resonant structure.

Equations (2), (3) and (4) show that the determination of the dielectric loss tangent of the sample requires calculations of volume and surface integrals. Alternatively, incremental frequency rules can be used for evaluation of the electric energy filling factor and geometric factor [12, 13]. Both direct integration and incremental frequency techniques give results that differ in the fourth decimal place so errors associated with such discrepancies can be neglected in determination of the loss tangent. Uncertainty in determination of the loss tangent employing dielectric resonator modes (modes having electric energy filling factors close to unity) depends primarily on two factors: parasitic losses in metal walls of the cavity and uncertainty of the unloaded Q -factor measurements. Results of computations of resonant frequencies, electric energy filling factors and geometric factors for the modes shown in figure 4 are shown in table 2. One can see that all modes shown in table 2 are dielectric resonator modes with electric energy filling factors $p_e > 0.98$. It is also seen that Q -factors due to conductor losses for higher order modes are substantially larger than for the dominant quasi TE₀₁₁ modes. This means that dielectric losses of low-loss materials can be more precisely measured by employing higher order modes rather than the dominant one.

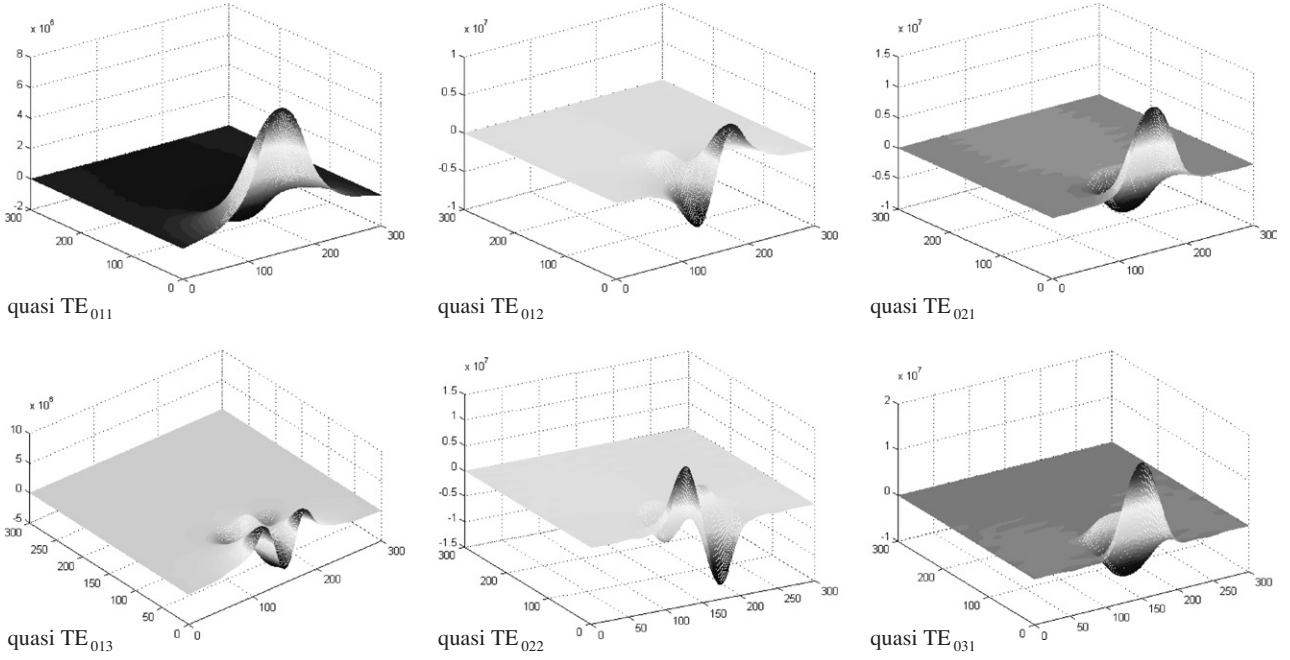


Figure 4. Distribution of the axial component of the magnetic field (Hz) for the first nine modes of the resonant structure having aspect ratio equal to 4 and $\epsilon_r = 100$. Cavity $D_c/L_c = 2$, $L_1 = (L_c - h)/2$.

3. Experiments

Experiments have been performed employing a network analyser (Agilent 8722 ES) and two silver-plated cavities having dimensions: $D_c = 25.0$ mm, $L_c = 12.12$ mm (larger cavity) and $D_c = 9.38$ mm, $L_c = 4.125$ mm (smaller cavity). Each cavity had been equipped with two adjustable horizontal coupling loops terminating semirigid coaxial cables that allow precise adjustment of coupling coefficients from both ports of the transmission resonator. The diameters of the cavities were measured with a precision of 0.01 mm and their lengths were evaluated from measured values of the TE_{011} mode resonant frequencies of the empty cavities and the diameters. Such a procedure allows one to neglect the influence of uncertainty of the cavity dimensions on the uncertainty of permittivity. Cylindrical dielectric supports have been made of oriented single-crystal quartz (cylinder axis parallel to crystal c -axis). The surface resistance of the cavity walls was evaluated for each cavity at a fixed frequency from the unloaded Q -factor measurements of empty cavities operating on the TE_{011} mode. To evaluate the loss tangent the surface resistance values were scaled proportionally to the square root of the frequency corresponding to the resonant frequency of a specific mode. In measurements of Q -factors coupling coefficients were adjusted to very small values ($S_{21} < -45$ dB) so the difference between loaded Q -factors, which were directly measured using the network analyser, and the unloaded Q -factors were negligible.

In order to perform accurate measurements of permittivity and dielectric loss tangent employing one of the higher order quasi TE_{0np} modes the following requirements should be satisfied:

- dielectric resonator modes (in contrast to the cavity modes) should be used,
- measured modes should be properly identified,

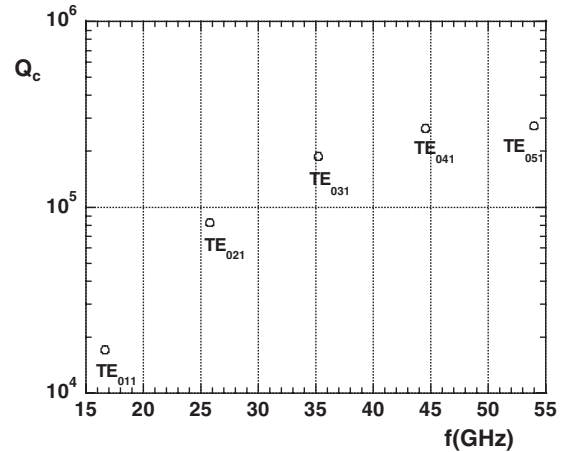


Figure 5. Q -factors due to conductor losses (Q_c) for the first five TE_{0n1} modes excited in a BMT sample having $d = 6.0$ mm, $h = 0.9$ mm, $\epsilon_r = 24.5$ situated centrally in a cavity with $D_c = 9.0$ mm $L_c = 4.1$ mm ($R_s = 30$ m Ω at 10 GHz).

- the contribution of conductor losses to the total Q -factor should be small.

The first requirement is in contradiction with the third one so one should take care to keep relative size of the sample (normalized to the size of the cavity) within a range that provides the best accuracy in determining the loss tangent.

For high permittivity materials mode identification becomes easier for samples that have a larger aspect ratio (d/h). For such samples the electromagnetic field distribution along the z -axis becomes similar for all modes belonging to the TE_{0n1} mode family. These modes are also less sensitive to the displacement of the sample position along the z -axis. In addition, for samples having larger aspect ratios the resonant frequencies of other TE_{0np} modes with $p > 1$ are shifted

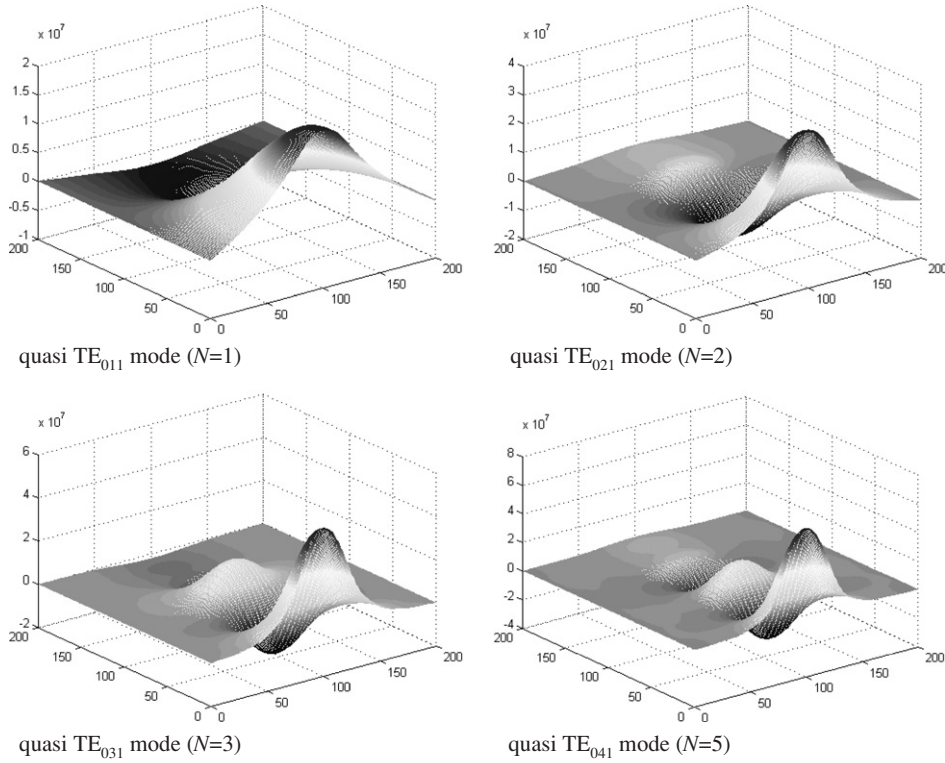


Figure 6. Distribution of the axial component of the magnetic field (Hz) for the first four TE_{0nl} modes excited in a BMT sample having $d = 6.0$ mm, $h = 0.9$ mm, $\epsilon_r = 24.5$ situated centrally in a cavity with $D_c = 9.0$ mm $L_c = 4.1$ mm.

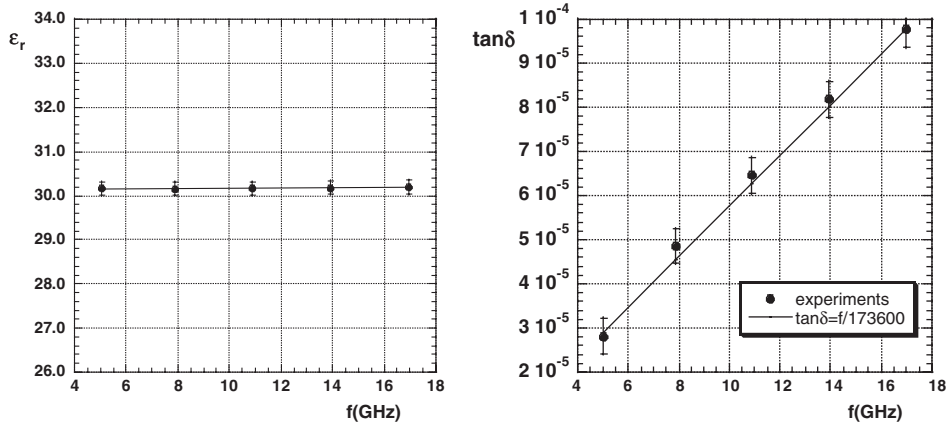


Figure 7. Permittivity and dielectric loss tangent of a BZT (Filtronics) sample having $d = 16.78$ mm and $h = 3.03$ mm in the cavity with $D_c = 25.0$ mm, $L_c = 12.12$ mm on a single-crystal quartz support with $d_s = 7.2$ mm, $d_{si} = 1.1$ mm, $L_1 = 4.02$ mm, $\epsilon_s = 4.43$, $\tan \delta_s = 2.2 \times 10^{-5}$, $R_s = 30$ m Ω at 10 GHz.

towards higher frequencies in the mode spectrum. Q -factors due to conductor losses for the TE_{0nl} mode family increase with increasing radial mode number, as is shown in figure 5 for the BMT ceramic sample used in our experiments. Magnetic field distributions for the first four TE_{0nl} modes are shown in figure 6.

In the experimental procedure modes were identified in the following way. Firstly, the dominant quasi TE_{011} mode was measured and the permittivity was evaluated. Then the resonant frequencies for a number of higher order TE_{0np} modes were computed. Finally the higher order modes were measured and the permittivity and loss tangent values were determined independently from the resonant frequencies and unloaded Q -factors of these modes.

The results of measurements for a few low-loss dielectric materials including ceramics and single crystals are presented in figures 7–10. One can notice that for all measured materials the permittivity is constant versus frequency and permittivity values obtained from different modes differ by no more than $\pm 0.5\%$. Small discrepancies are partly due to dimensional uncertainties and partly due to computational errors. In order to obtain such measurement uncertainties of permittivity dimensional uncertainties must be below 0.2% and computational uncertainties below 0.1%, which has been satisfied for all our samples.

In practice, for most measured materials except single-crystal quartz, the dielectric loss tangents increase almost linearly with frequency so the approximation $f/\tan \delta =$

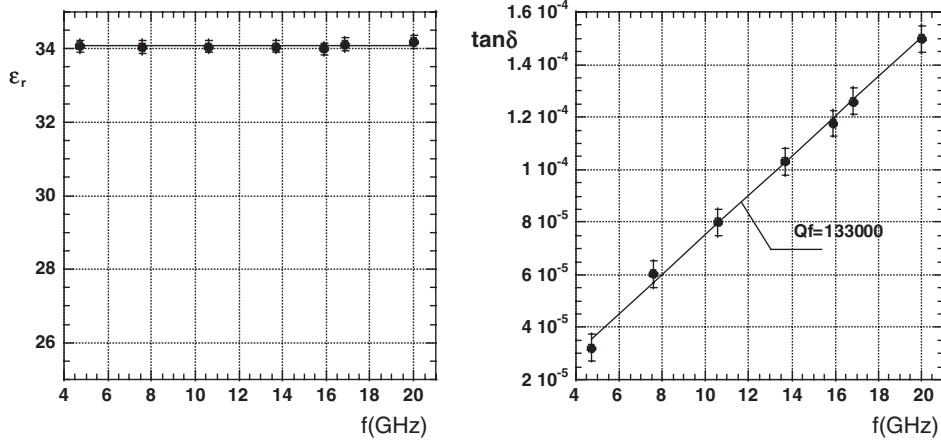


Figure 8. Permittivity and dielectric loss tangent of a BZNT (Filtronics) sample having $d = 15.43$ mm, $h = 3.99$ measured in a cavity the cavity with $D_c = 20.02$ mm, $L_c = 12.12$ mm on a single-crystal quartz support with $d_s = 7.99$ mm, $d_{si} = 1.1$ mm, $L_1 = 4.48$ mm, $\varepsilon_s = 4.43$, $\tan \delta_s = 2.2 \times 10^{-5}$, $R_s = 30$ m Ω at 10 GHz.

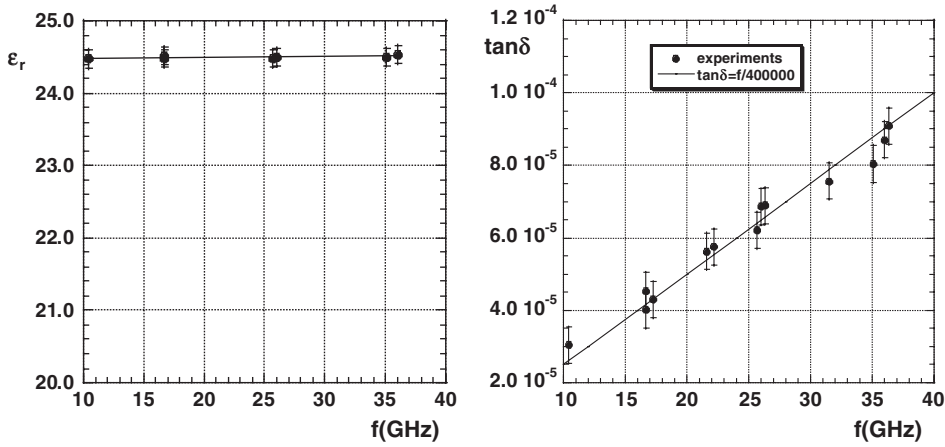


Figure 9. Permittivity and dielectric loss tangent of a BMT ceramic (Neomax). Results have been obtained in two different cavities for three different samples having $d = 6.0$ mm, and heights $h = 0.9$ mm, $h = 1.8$ mm (two samples stacked together) and $h = 2.7$ mm (three samples stacked together).

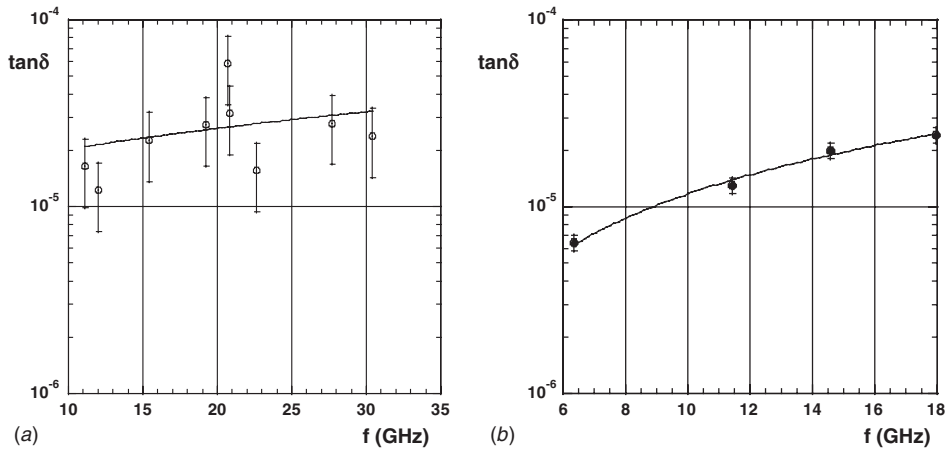


Figure 10. Dielectric loss tangents perpendicular to the anisotropy axis for single-crystal quartz (a) (measured on three different samples) and single-crystal sapphire (b) (measured on two different samples). The measured permittivity was $4.43 \pm 0.5\%$ for single-crystal quartz and $9.39 \pm 0.5\%$ for sapphire.

constant is well satisfied. Uncertainties in dielectric loss tangent for high permittivity materials are typically smaller than $\pm 5\%$. For low permittivity materials, such as single-

crystal quartz, they become larger as one can see in figure 10. The main reason for larger uncertainties in the dielectric loss tangent for low permittivity materials is a relatively high

contribution of conductor losses in the measured Q -factor value. Results of experiments obtained in this paper for high purity single crystals are in excellent agreement with results obtained from measurements employing whispering gallery modes [14].

4. Conclusions

It has been proved both theoretically and experimentally that higher order quasi modes can be used for precise measurements of permittivity and dielectric loss tangents of dielectric materials versus frequency. Using this technique, dielectrics having arbitrary permittivity values can be measured with uncertainties better than $\pm 0.5\%$. To obtain such uncertainties the dimensions of samples under test must be measured with a precision better than 0.2% . For high permittivity dielectrics ($\epsilon_r > 9$) losses as low as 10^{-6} can be precisely measured; for low permittivity dielectrics ($\epsilon_r < 4$) loss tangent values of the order 10^{-5} are measurable. The maximum losses that can be measured using this technique are limited by the minimum measurable Q -factor value, which is of the order of 100. As a result, the maximum measurable dielectric losses are of the order of 10^{-2} .

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